A Mini Project Report of

PCC-201-COM : Data Structures

Submitted by

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[2026-27]

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CERTIFICATE

This is to certify that the "**Mini Project Report**" submitted by **Purva Pramod Raut (30),** on **"Implementation of a basic version of a web browser's back button functionality [Stack]"** is work done by her and submitted during 2026–2027 academic year, in partial fulfillment of the requirements for the award of the degree of **Bachelor of engineering** in computer Engineering, at **Bharati Vidyapeeth's College of Engineering, Lavale, Pune**.

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ACKNOWLEDGEMENT

It gives immense pleasure to express our gratitude & thanks to all those who have helped in the timely completion of the Mini-Project work. We have a great pleasure in expressing our deep sense of gratitude and indebtedness to **Prof. Tejal H. Patil** Guide for their invaluable guidance, constant supervision and sustained encouragement throughout the tenure in this Mini Project work.

We would like to express our thanks to **Dr. Lalit V. Patil**, Head of Department, Department of Computer Engineering Bharati Vidyapeeth's College of Engineering, Lavale, Pune for their constant support and encouragement.

We would like to express our thanks to **Prof. Dr. R. N. Patil** Principal, Bharati Vidyapeeth's College of Engineering, Lavale, Pune for permitting us to take up the project work. We are thankful to our parents and friends for their kind help and support throughout the course.

Purva Pramod Raut
Student Name

ABSTRACT

The project “Implementation of a Basic Version of a Web Browser’s Back Button Functionality Using Stack” aims to demonstrate the practical application of the stack data structure in web browsing. A web browser allows users to visit different web pages and return to previously visited pages using the Back button. This functionality can be implemented efficiently using the Last In, First Out (LIFO) principle of a stack.

In this project, each visited web page is stored in a stack. Whenever a user navigates to a new page, the current page is pushed onto the stack. When the user clicks the Back button, the most recently visited page is retrieved by popping the top element from the stack. This process helps users navigate backward through their browsing history in reverse order.

The program also checks whether the stack is empty before performing a pop operation, preventing errors when the user tries to go back beyond the available browsing history. This project demonstrates how stack operations such as push and pop can be used to implement simple browser navigation. It provides a clear understanding of stack data structures and their practical applications in computer science.

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INTRODUCTION

Overview:

A web browser is a software application used to access and view web pages on the internet. While browsing, users often visit multiple web pages and need to return to previously visited pages. The Back button helps users navigate to the previous page without manually entering its address again.

The stack is a linear data structure that follows the Last In, First Out (LIFO) principle. This means the element inserted last is removed first. Stack operations such as push and pop can be used to implement the basic functionality of a browser's Back button.

In this project, a stack is used to store the browsing history. Whenever a user visits a new page, the current page is added to the stack. When the Back button is pressed, the most recently visited page is retrieved from the stack. This project explains the practical use of stacks in browser navigation and helps develop a clear understanding of stack operations.

Objective:

1. To understand the concept and working of the stack data structure.
2. To implement the LIFO principle in a practical application.
3. To store visited web pages using the push operation.
4. To navigate to the previously visited page using the pop operation.
5. To handle an empty stack condition when no previous page is available.
6. To demonstrate the practical application of stacks in web browser navigation.

LITERATURE SURVEY

Year	Paper Name	Author	Conclusion
1999–2000	Getting Back to Back: Alternate Behaviors for a Web Browser's Back Button	Saul Greenberg and Andy Cockburn	The paper explains how the stack-based Back button helps users revisit recently visited web pages and discusses its advantages and limitations.
2002	Pushing Back: Evaluating a New Behaviour for the Back and Forward Buttons in Web Browsers	Andy Cockburn, Bruce McKenzie and Michael JasonSmith	The study compares the traditional stack-based approach with an alternative that maintains a complete list of previously visited pages.
2008	Back vs. Stack: Training the Correct Mental Model Affects Web Browsing	M. Uther and H. Haley	The research examines users' understanding of the browser's Back button and how understanding its behavior affects web browsing.
2012	Contrasting Stack-Based and Recency-Based Back Buttons on Web Browsers	Saul Greenberg, Geoffrey Ho and Shaun Kaasten	The paper compares stack-based and recency-based navigation and discusses the advantages and limitations of each approach.

UML DIAGRAM

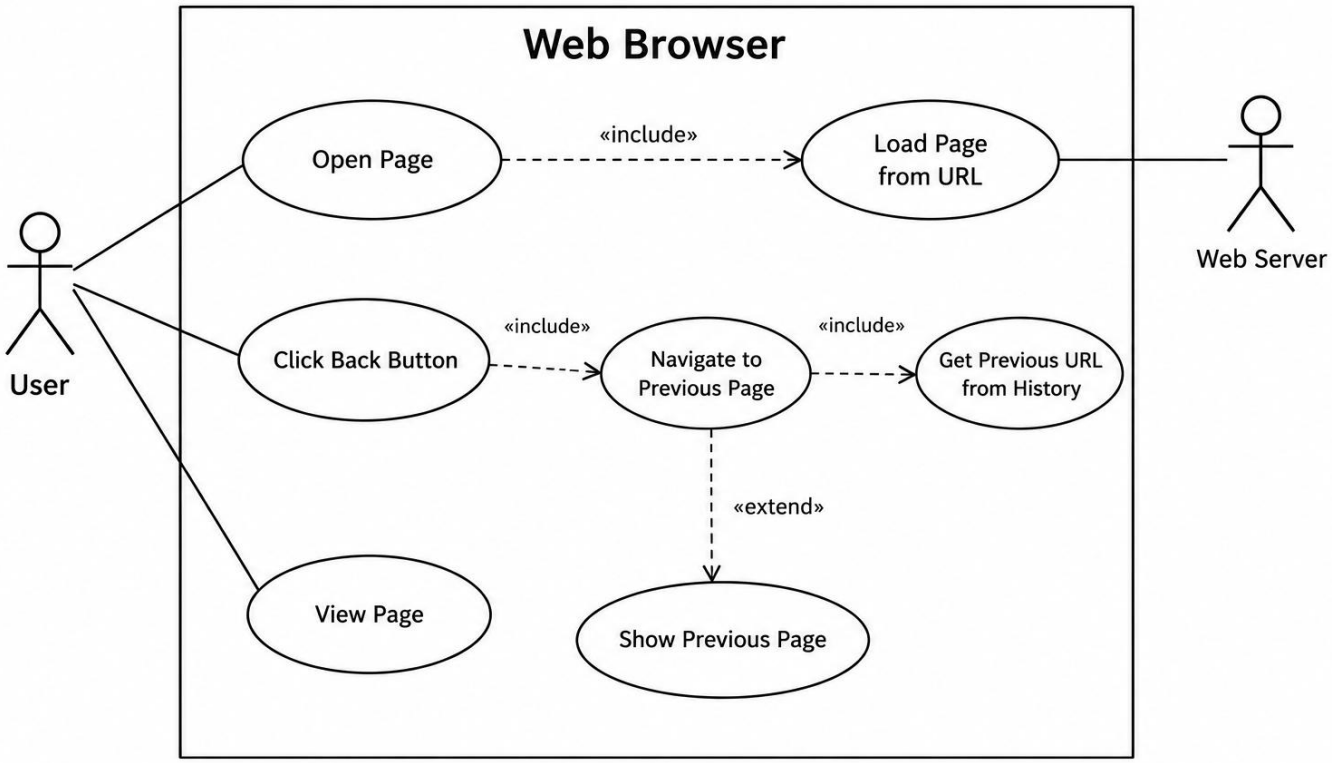


Fig 3.1: UML Diagram

SYSTEM REQUIREMENT

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Hard Disk: Minimum 500 MB free space
- Input Device: Keyboard and mouse
- Output Device: Monitor

Software Requirements

- Operating System: Windows 10/11 or Linux
- Programming Language: Java / Python / C++
- IDE: Visual Studio Code, Code::Blocks, or any suitable IDE
- Data Structure: Stack
- Compiler/Interpreter: Appropriate compiler or interpreter for the selected programming language

SYSTEM DESIGN

Exixting System

In a traditional web browsing system, users navigate between different web pages to access information. The browser's back button allows users to return to previously visited pages. However, implementing this functionality requires maintaining a record of browsing history in the correct order. Without a proper data structure, managing previously visited pages can become difficult.

The existing system has the following limitations:

1. **Difficult History Management:** Managing previously visited web pages without a suitable data structure can be complicated.
2. **Time-Consuming Navigation:** Finding and returning to a previously visited page can be inconvenient if browsing history is not maintained properly.
3. **Incorrect Navigation Order:** Without an organized mechanism, returning to the most recently visited page may become difficult.
4. **Limited Efficiency:** Managing browsing history becomes more complex as the number of visited pages increases.

Proposed System

The proposed Web Browser Back Button System is a Python-based, menu-driven application developed using the Stack data structure. It maintains a history of visited web pages and allows users to navigate backward through previously visited pages.

The proposed system provides the following features:

1. **Web Page Navigation:** Allows users to visit new web pages by entering their URLs or page names.
2. **Stack-Based History Management:** Uses the Stack data structure to maintain the browsing history in the order of visits.
3. **Back Button Functionality:** Uses the pop() operation to return to the previously visited page.
4. **Current Page Display:** Displays the currently active web page after each navigation operation.
5. **Browsing History Management:** Displays the previously visited pages stored in the stack.

Entity Relationship Diagram:

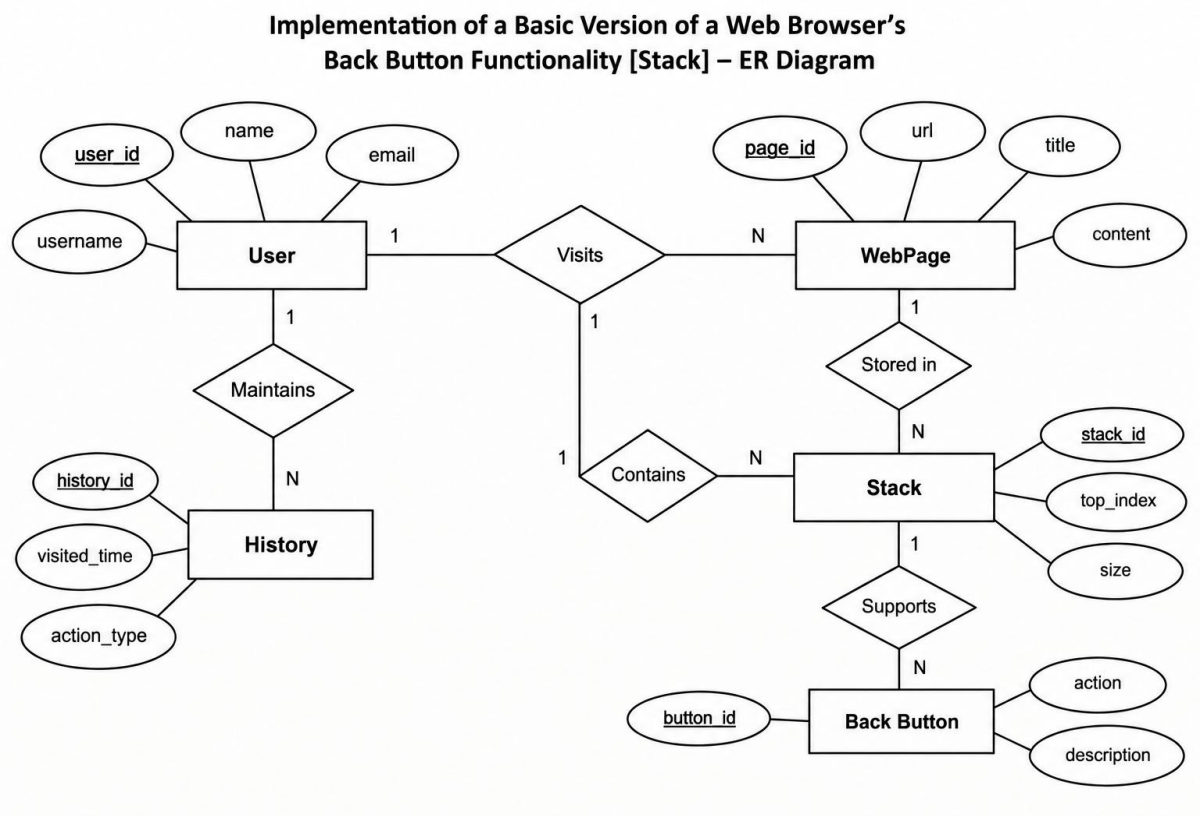


Fig 5.1: ER Diagram

SYSTEM CODE

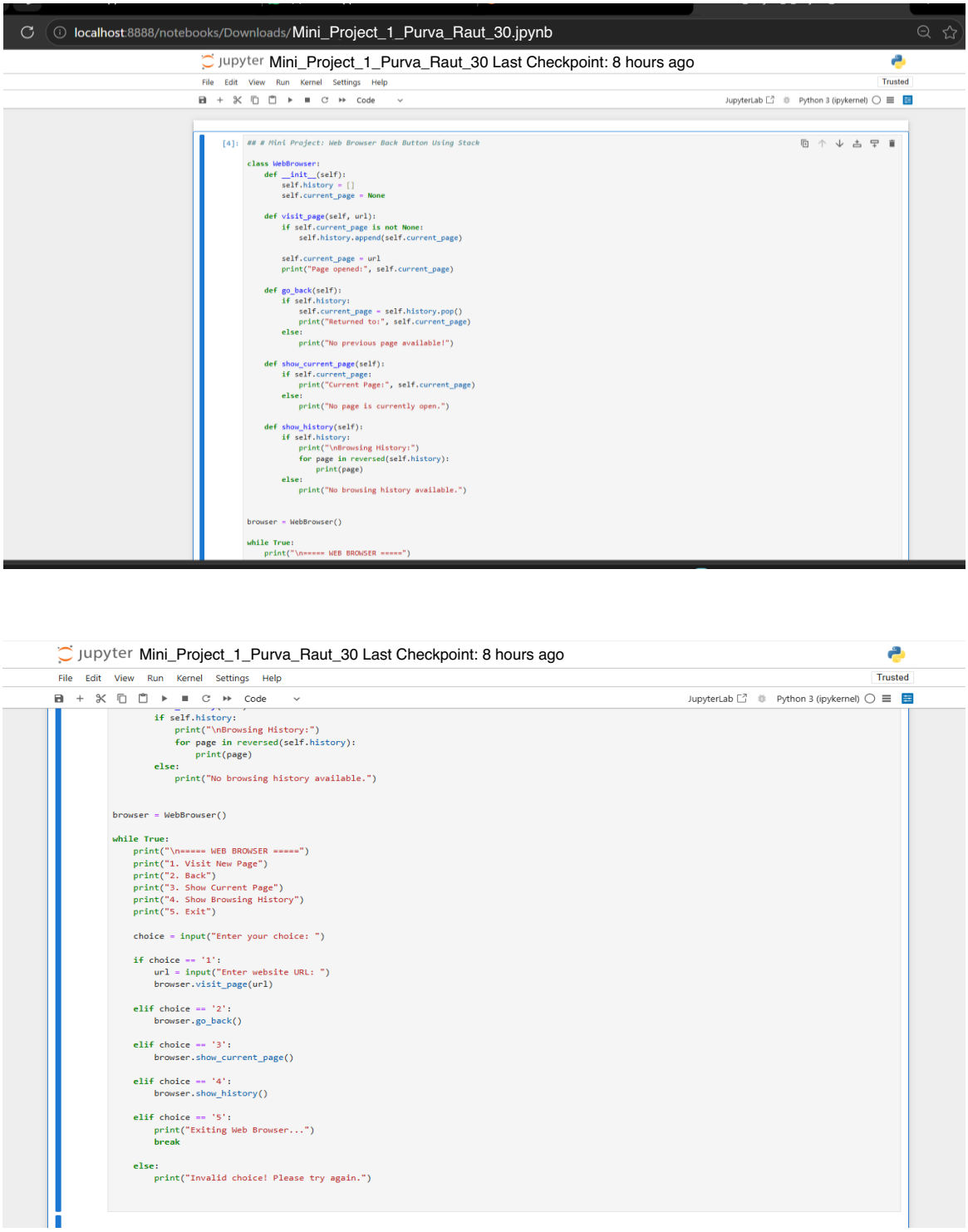
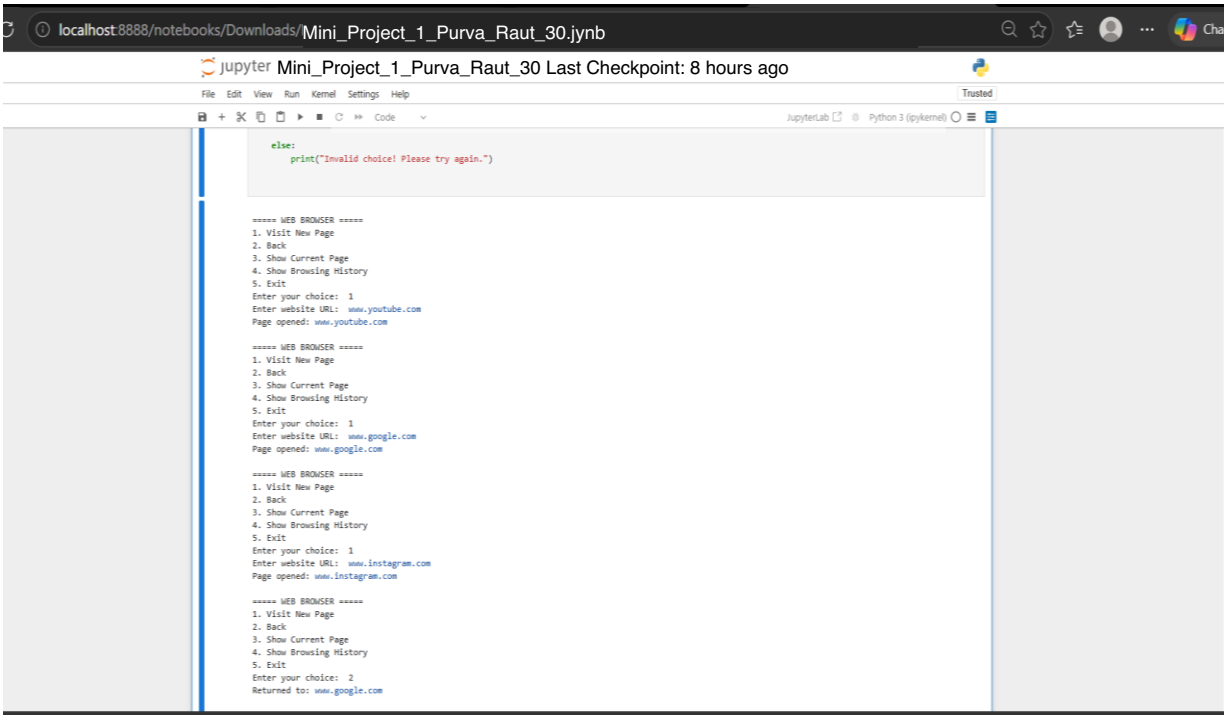


Fig 5.2: System code

RESULT



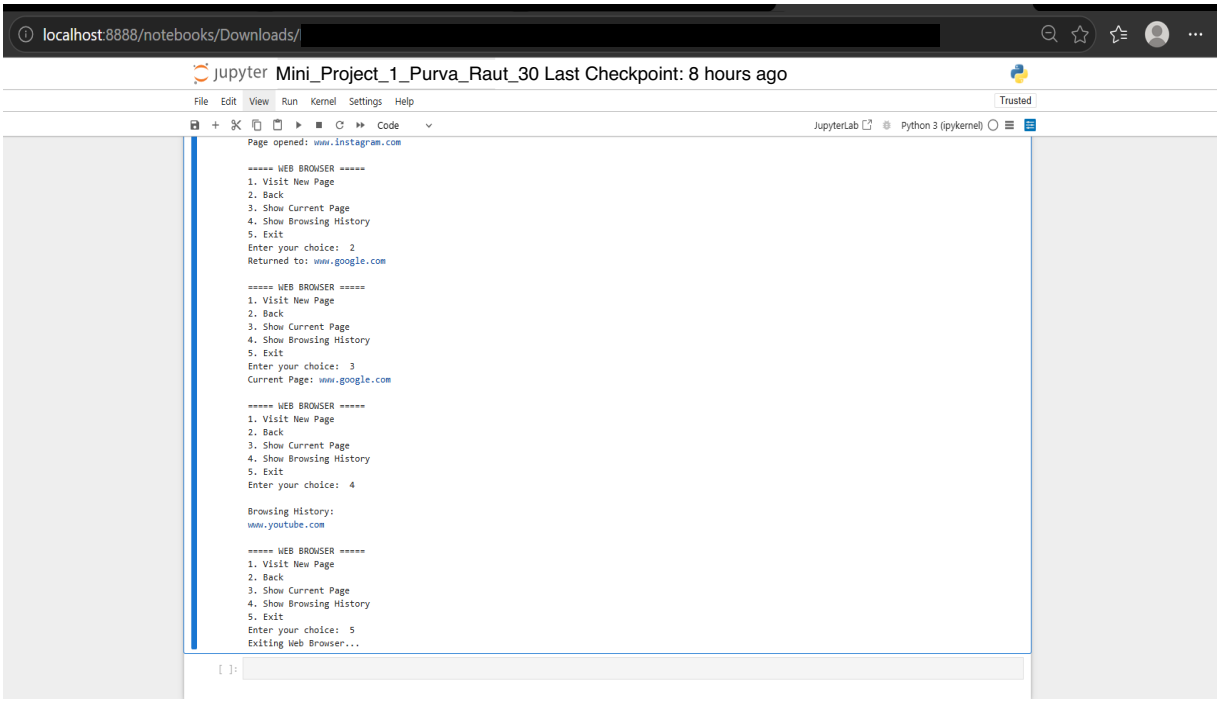


Fig 5.3 Result

APPLICATIONS

1. Web Browsers: Used to implement the Back button to return to previously visited web pages.
2. Mobile Applications: Helps users navigate back to previously viewed screens.
3. Undo Operations: Stack concepts are used to reverse the most recent actions in text editors and other applications.
4. Navigation Systems: Useful for returning to previously visited pages or menu screens.
5. Software Development: Helps programmers understand the practical implementation of stack data structures.
6. History Management: Used to maintain and retrieve recently visited pages in applications.
7. Menu Navigation: Helps users return to the previous menu or screen in software applications.

FUTURE WORK

The basic web browser Back button functionality using a stack can be improved in several ways in the future.

1. Forward Button Functionality: Add a Forward button to allow users to move to the next page after going back.
2. Browsing History: Maintain a complete list of visited web pages with timestamps.
3. Graphical User Interface (GUI): Develop a user-friendly interface with Back and Forward buttons.
4. Multiple Tabs: Support multiple browser tabs, each with its own browsing history.
5. History Search: Allow users to search for previously visited web pages.
6. Data Storage: Store browsing history permanently so users can access it even after restarting the application.
7. Improved Error Handling: Provide suitable messages when no previous page is available.

These improvements can make the system more practical, user-friendly, and similar to a real web browser.

CONCLUSION

The project "Implementation of a Basic Version of a Web Browser's Back Button Functionality Using Stack" demonstrates the practical application of the stack data structure in web navigation. The system uses the Last In, First Out (LIFO) principle to store previously visited web pages and retrieve them when the Back button is pressed.

By implementing push and pop operations, the project provides a simple and efficient method for managing browsing history. It also handles the empty-stack condition to prevent invalid operations. Through this project, we gain a better understanding of stack operations and their real-world applications in computer science.

In conclusion, the project successfully explains how a stack can be used to implement basic browser Back button functionality. Further improvements, such as adding a Forward button, multiple tabs, and permanent browsing history, can make the system more advanced and useful.

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